COPY Command:

- **Basic COPY Syntax:**

```
lab_17=# COPY employees TO 'C:/Users/Public/lab17/employees.csv' WITH CSV HEADER;
COPY 10
lab_17=# COPY employees_import FROM 'C:/Users/Public/lab17/employees.csv' WITH CSV HEADER;
COPY 10
```

- **Advanced COPY options:**

```
lab_17=# COPY employees TO 'C:/Users/Public/lab17/employees.txt'
lab_17-# WITH DELIMITER '|' NULL 'N/A' CSV HEADER;
COPY 10
lab_17=# COPY employees(first_name, last_name, email)
lab_17-# FROM 'C:/Users/Public/lab17/new_employees.csv' WITH CSV HEADER;
COPY 3
lab_17=# COPY (SELECT * FROM employees WHERE department = 'IT')
lab_17-# TO 'C:/Users/Public/lab17/it_employees.csv' WITH CSV HEADER;
COPY 4
```

CSV Import/Export:

- **CSV Export:**

```
lab_17=# COPY products TO 'C:/Users/Public/lab17/products.csv' WITH CSV HEADER;
COPY 10
lab_17=# COPY customers TO 'C:/Users/Public/lab17/customers.csv'
lab_17-# WITH CSV HEADER DELIMITER ';' QUOTE '"';
COPY 10
lab_17=# COPY orders TO 'C:/Users/Public/lab17/orders.csv'
lab_17-# WITH CSV HEADER FORCE QUOTE order_date;
COPY 15
```

- **CSV Import:**

```
lab_17=# COPY products FROM 'C:/Users/Public/lab17/products.csv' WITH CSV HEADER;
COPY 10
lab_17=# COPY products FROM 'C:/Users/Public/lab17/products_with_errors.csv'
lab_17-# WITH (FORMAT csv, HEADER, ON_ERROR ignore);
NOTICE:  2 rows were skipped due to data type incompatibility
COPY 3
lab_17=# COPY products FROM 'C:/Users/Public/lab17/products_utf8.csv'
lab_17-# WITH CSV HEADER ENCODING 'UTF8';
COPY 3
```

- **Handling special cases:**

```
lab_17=# COPY customer_feedback TO 'C:/Users/Public/lab17/feedback.csv'
lab_17-# WITH CSV HEADER DELIMITER ',' QUOTE '"' ESCAPE '"';
COPY 6
lab_17=# COPY customer_feedback TO 'C:/Users/Public/lab17/feedback.csv'
lab_17-# WITH CSV HEADER DELIMITER ',' QUOTE '"' ESCAPE '"';
COPY 6
lab_17=# COPY sales_data FROM 'C:/Users/Public/lab17/sales.csv'
lab_17-# WITH CSV HEADER NULL 'NULL';
COPY 5
```

pg_dump - Creating Backups:

```
C:\Users\Public\lab17>pg_dump -h localhost -U postgres -d lab_17 > backup.sql

C:\Users\Public\lab17>pg_dump -h localhost -U postgres -d lab_17 -Fc > backup.dump

C:\Users\Public\lab17>pg_dump -h localhost -U postgres -d lab_17 -t employees -t departments > tables_backup.sql

C:\Users\Public\lab17>pg_dump -h localhost -U postgres -d lab_17 -Fc -v > backup.dump
pg_dump: executing SELECT pg_catalog.set_config('search_path', '', false);
pg_dump: last built-in OID is 16383
pg_dump: reading extensions
pg_dump: identifying extension members
pg_dump: reading schemas
pg_dump: reading user-defined tables
pg_dump: reading user-defined functions
pg_dump: reading user-defined types
pg_dump: reading procedural languages
pg_dump: reading user-defined aggregate functions
pg_dump: reading user-defined operators
pg_dump: reading user-defined access methods
pg_dump: reading user-defined operator classes
pg_dump: reading user-defined operator families
pg_dump: reading user-defined text search parsers
pg_dump: reading user-defined text search templates
pg_dump: reading user-defined text search dictionaries
pg_dump: reading user-defined text search configurations
pg_dump: reading user-defined foreign-data wrappers
pg_dump: reading user-defined foreign servers
pg_dump: reading default privileges
pg_dump: reading user-defined collations
pg_dump: reading user-defined conversions
pg_dump: reading type casts
pg_dump: reading transforms
pg_dump: reading table inheritance information
pg_dump: reading event triggers
pg_dump: finding extension tables
pg_dump: finding inheritance relationships
pg_dump: reading column info for interesting tables
pg_dump: finding table default expressions
pg_dump: flagging inherited columns in subtables
pg_dump: reading partitioning data
pg_dump: reading indexes
pg_dump: flagging indexes in partitioned tables
pg_dump: reading extended statistics
```

pg restore - Restoring Backups:

```
C:\Users\Public\lab17>pg_restore -h localhost -U postgres -d target_database backup.dump

C:\Users\Public\lab17>createdb -h localhost -U postgres new_database

C:\Users\Public\lab17>pg_restore -h localhost -U postgres -d new_database backup.dump

C:\Users\Public\lab17>createdb -h localhost -U postgres tables_restore

C:\Users\Public\lab17>pg_restore -h localhost -U postgres -d tables_restore -t employees backup.dump

C:\Users\Public\lab17>createdb -h localhost -U postgres parallel_restore

C:\Users\Public\lab17>pg_restore -h localhost -U postgres -d parallel_restore -j 4 backup.dump
```

Advanced pg_dump options:

```
C:\Users\Public\lab17>pg_dump -h localhost -U postgres -d lab_17 -s > schema_only.sql

C:\Users\Public\lab17>pg_dump -h localhost -U postgres -d lab_17 -a > data_only.sql

C:\Users\Public\lab17>pg_dump -h localhost -U postgres -d lab_17 -T log_table -T temp_data > backup_without_logs.sql
```

Full Database Backups:

- **Creating full backups:**

```
C:\Users\Public\lab17>pg_dumpall -h localhost -U postgres > full_cluster_backup.sql

C:\Users\Public\lab17>pg_dump -h localhost -U postgres -d lab_17 -Fc --verbose > full_production_backup.dump
pg_dump: executing SELECT pg_catalog.set_config('search_path', '', false);
pg_dump: last built-in OID is 16383
pg_dump: reading extensions
pg_dump: identifying extension members
pg_dump: reading schemas
pg_dump: reading user-defined tables
pg_dump: reading user-defined functions
pg_dump: reading user-defined types
pg_dump: reading procedural languages
pg_dump: reading user-defined aggregate functions
pg_dump: reading user-defined operators
pg_dump: reading user-defined access methods
pg_dump: reading user-defined operator classes
pg_dump: reading user-defined operator families
pg_dump: reading user-defined text search parsers
pg_dump: reading user-defined text search templates
pg_dump: reading user-defined text search dictionaries
pg_dump: reading user-defined text search configurations
pg_dump: reading user-defined foreign-data wrappers
pg_dump: reading user-defined foreign servers
pg_dump: reading default privileges
pg_dump: reading user-defined collations
pg_dump: reading user-defined conversions
pg_dump: reading type casts
pg_dump: reading transforms
pg_dump: reading table inheritance information
pg_dump: reading event triggers
pg_dump: finding extension tables
pg_dump: finding inheritance relationships
pg_dump: reading column info for interesting tables
pg_dump: finding table default expressions
pg_dump: flagging inherited columns in subtables
pg_dump: reading partitioning data
pg_dump: reading indexes
pg_dump: flagging indexes in partitioned tables
pg_dump: reading extended statistics
pg_dump: reading constraints
pg_dump: reading triggers
pg_dump: reading rewrite rules
pg_dump: reading policies
pg_dump: reading row-level security policies
```

```
pg_dump: reading policies
pg_dump: reading row-level security policies
pg_dump: reading publications
pg_dump: reading publication membership of tables
pg_dump: reading publication membership of schemas
pg_dump: reading subscriptions
pg_dump: reading subscription membership of tables
pg_dump: reading large objects
pg_dump: reading dependency data
pg_dump: saving encoding = UTF8
pg_dump: saving "standard_conforming_strings = on"
pg_dump: saving "search_path = "
pg_dump: saving database definition
pg_dump: dumping contents of table "public.customer_feedback"
pg_dump: dumping contents of table "public.customers"
pg_dump: dumping contents of table "public.departments"
pg_dump: dumping contents of table "public.employees"
pg_dump: dumping contents of table "public.employees_import"
pg_dump: dumping contents of table "public.large_table_2023"
pg_dump: dumping contents of table "public.log_table"
pg_dump: dumping contents of table "public.orders"
pg_dump: dumping contents of table "public.products"
pg_dump: dumping contents of table "public.sales_data"
pg_dump: dumping contents of table "public.staging_customers"
pg_dump: dumping contents of table "public.target_customers"
pg_dump: dumping contents of table "public.temp_data"

C:\Users\Public\lab17>pg_dump -h localhost -U postgres -d lab_17 -Fc -O > backup_with_ownership.dump
```

- **Backup Validation:**

```
C:\Users\Public\lab17>pg_restore --list backup.dump
;
; Archive created at 2026-10-27 10:18:05
;     dbname: lab_17
;     TOC Entries: 81
;     Compression: gzip
;     Dump Version: 1.16-0
;     Format: CUSTOM
;     Integer: 4 bytes
;     Offset: 8 bytes
;     Dumped from database version: 18.6
;     Dumped by pg_dump version: 18.6
;
;
; Selected TOC Entries:
;
231; 1259 17962 TABLE public customer_feedback postgres
230; 1259 17961 SEQUENCE public customer_feedback_feedback_id_seq postgres
5024; 0 0 SEQUENCE OWNED BY public customer_feedback_feedback_id_seq postgres
227; 1259 17941 TABLE public customers postgres
226; 1259 17940 SEQUENCE public customers_customer_id_seq postgres5025; 0 0 SEQUENCE OWNED BY public customers_customer_id_seq postgres
220; 1259 17911 TABLE public departments postgres
219; 1259 17910 SEQUENCE public departments_department_id_seq postgres
5026; 0 0 SEQUENCE OWNED BY public departments_department_id_seq postgres
222; 1259 17919 TABLE public employees postgres
221; 1259 17918 SEQUENCE public employees_employee_id_seq postgres5027; 0 0 SEQUENCE OWNED BY public employees_employee_id_seq postgres
223; 1259 17926 TABLE public employees_import postgres
239; 1259 18006 TABLE public large_table postgres
238; 1259 18005 SEQUENCE public large_table_id_seq postgres
5028; 0 0 SEQUENCE OWNED BY public large_table_id_seq postgres
241; 1259 18018 TABLE public large_table_2023 postgres
234; 1259 17980 TABLE public log_table postgres
233; 1259 17979 SEQUENCE public log_table_log_id_seq postgres
5029; 0 0 SEQUENCE OWNED BY public log_table_log_id_seq postgres
229; 1259 17949 TABLE public orders postgres
228; 1259 17948 SEQUENCE public orders_order_id_seq postgres
5030; 0 0 SEQUENCE OWNED BY public orders_order_id_seq postgres
225; 1259 17933 TABLE public products postgres
224; 1259 17932 SEQUENCE public products_product_id_seq postgres
5031; 0 0 SEQUENCE OWNED BY public products_product_id_seq postgres
232; 1259 17976 TABLE public sales_data postgres
240; 1259 18015 TABLE public staging_customers postgres
237; 1259 17999 TABLE public target_customers postgres
236; 1259 17990 TABLE public temp_data postgres
```

```
232; 1259 17976 TABLE public sales_data postgres
240; 1259 18015 TABLE public staging_customers postgres
237; 1259 17999 TABLE public target_customers postgres
236; 1259 17990 TABLE public temp_data postgres
235; 1259 17989 SEQUENCE public temp_data_id_seq postgres
5032; 0 0 SEQUENCE OWNED BY public temp_data_id_seq postgres
4815; 0 0 TABLE ATTACH public large_table_2023 postgres
4821; 2604 17965 DEFAULT public customer_feedback feedback_id postgres
4819; 2604 17944 DEFAULT public customers customer_id postgres
4816; 2604 17914 DEFAULT public departments department_id postgres4817; 2604 17922 DEFAULT public employees employee_id postgres
4824; 2604 18009 DEFAULT public large_table id postgres
4822; 2604 17983 DEFAULT public log_table log_id postgres
4820; 2604 17952 DEFAULT public orders order_id postgres
4818; 2604 17936 DEFAULT public products product_id postgres
4823; 2604 17993 DEFAULT public temp_data id postgres
5008; 0 17962 TABLE DATA public customer_feedback postgres
5004; 0 17941 TABLE DATA public customers postgres
4997; 0 17911 TABLE DATA public departments postgres
4999; 0 17919 TABLE DATA public employees postgres
5000; 0 17926 TABLE DATA public employees_import postgres
5017; 0 18018 TABLE DATA public large_table_2023 postgres
5011; 0 17980 TABLE DATA public log_table postgres
5006; 0 17949 TABLE DATA public orders postgres
5002; 0 17933 TABLE DATA public products postgres
5009; 0 17976 TABLE DATA public sales_data postgres
5016; 0 18015 TABLE DATA public staging_customers postgres
5014; 0 17999 TABLE DATA public target_customers postgres
5013; 0 17990 TABLE DATA public temp_data postgres
5033; 0 0 SEQUENCE SET public customer_feedback_feedback_id_seq postgres
5034; 0 0 SEQUENCE SET public customers_customer_id_seq postgres
5035; 0 0 SEQUENCE SET public departments_department_id_seq postgres
5036; 0 0 SEQUENCE SET public employees_employee_id_seq postgres
5037; 0 0 SEQUENCE SET public large_table_id_seq postgres
5038; 0 0 SEQUENCE SET public log_table_log_id_seq postgres
5039; 0 0 SEQUENCE SET public orders_order_id_seq postgres
5040; 0 0 SEQUENCE SET public products_product_id_seq postgres
5041; 0 0 SEQUENCE SET public temp_data_id_seq postgres
4839; 2606 17970 CONSTRAINT public customer_feedback customer_feedback_pkey postgres
4835; 2606 17947 CONSTRAINT public customers customers_pkey postgres
4827; 2606 17917 CONSTRAINT public departments departments_pkey postgres
4831; 2606 17931 CONSTRAINT public employees_import employees_import_pkey postgres
4829; 2606 17925 CONSTRAINT public employees employees_pkey postgres
4841; 2606 17988 CONSTRAINT public log_table log_table_pkey postgres
4837; 2606 17955 CONSTRAINT public orders orders_pkey postgres
```

```
4837; 2606 17955 CONSTRAINT public orders orders_pkey postgres
4833; 2606 17939 CONSTRAINT public products products_pkey postgres4845; 2606 18004 CONSTRAINT public target_customers target_customers_pkey postgres
4843; 2606 17998 CONSTRAINT public temp_data temp_data_pkey postgres
4847; 2606 17971 FK CONSTRAINT public customer_feedback customer_feedback_customer_id_fkey postgres
4846; 2606 17956 FK CONSTRAINT public orders orders_customer_id_fkey postgres
4995; 6104 18014 PUBLICATION - migration_pub postgres

C:\Users\Public\lab17>createdb -h localhost -U postgres test_restore

C:\Users\Public\lab17>pg_restore -h localhost -U postgres -d test_restore backup.dump
```

- **Incremental Backups:**

```
lab_17=# SHOW wal_level;
 wal_level
-----------
 replica
(1 row)


lab_17=# SHOW archive_mode;
 archive_mode
--------------
 off
(1 row)


lab_17=# SHOW archive_command;
 archive_command
-----------------
 (disabled)
(1 row)


lab_17=# SELECT pg_switch_wal();
 pg_switch_wal
---------------
 0/6FD8018
(1 row)


lab_17=# SELECT pg_create_restore_point('before_data_migration');
 pg_create_restore_point
-------------------------
 0/7000090
(1 row)
```

Migration:

```
C:\Users\Public\lab17>pg_dump -h localhost -U postgres -d lab_17 -Fc > migration.dump

C:\Users\Public\lab17>createdb -h localhost -U postgres lab_17_target

C:\Users\Public\lab17>pg_restore -h localhost -U postgres -d lab_17_target migration.dump

C:\Users\Public\lab17>createdb -h localhost -U postgres lab_17_small

C:\Users\Public\lab17>pg_dump -h localhost -U postgres -d lab_17 -t departments | psql -h localhost -U postgres -d lab_17_small
SET
SET
SET
SET
SET
SET
 set_config
------------

(1 row)

SET
SET
SET
SET
SET
SET
CREATE TABLE
ALTER TABLE
CREATE SEQUENCE
ALTER SEQUENCE
ALTER SEQUENCE
ALTER TABLE
COPY 5
 setval
--------
      5
(1 row)

ALTER TABLE
```

Result:

```
lab_17-# HEADER;
COPY 10
lab_17=# INSERT INTO target_customers
lab_17-# SELECT * FROM staging_customers
lab_17-# ON CONFLICT (customer_id) DO UPDATE SET
lab_17-# name = EXCLUDED.name, email = EXCLUDED.email, city = EXCLUDED.city;
INSERT 0 10
lab_17=# SELECT pid, usename, application_name, client_addr, state
lab_17-# FROM pg_stat_activity
lab_17-# WHERE application_name = 'pg_dump';
 pid | usename | application_name | client_addr | state
-----+---------+------------------+-------------+-------
(0 rows)


lab_17=# SELECT COUNT(*) FROM target_customers;
 count
-------
    10
(1 row)


lab_17=# CREATE TABLE large_table_2023 PARTITION OF large_table
lab_17-# FOR VALUES FROM ('2023-01-01') TO ('2024-01-01');
CREATE TABLE
lab_17=# INSERT INTO large_table (created_at, amount) VALUES
lab_17-# ('2023-02-01', 100.00), ('2023-05-17', 250.50), ('2023-09-30', 75.25), ('2023-12-2
4', 310.00);
INSERT 0 4
```

```
C:\Users\Public\lab17>createdb -h localhost -U postgres lab_17_large

C:\Users\Public\lab17>psql -h localhost -U postgres -d lab_17_large -c "CREATE TABLE large_table (id SERIAL, created_at DATE NOT NULL, amount NUMERIC(10,2)) PARTITION B
Y RANGE (created_at);"
CREATE TABLE

C:\Users\Public\lab17>pg_dump -h localhost -U postgres -d lab_17 -t large_table_2023 | psql -h localhost -U postgres -d lab_17_large
SET
SET
SET
SET
SET
SET
 set_config
------------

(1 row)

SET
SET
SET
SET
SET
SET
CREATE TABLE
ALTER TABLE
ALTER TABLE
COPY 4
```

```
C:\Users\Public\lab17>dir
 Том в устройстве C имеет метку OS
 Серийный номер тома: 4761-1B74

 Содержимое папки C:\Users\Public\lab17

10/27/2026  10:28 AM    <DIR>          .
10/27/2026  10:28 AM    <DIR>          ..
10/27/2026  10:18 AM            26,307 backup.dump
10/27/2026  10:17 AM            22,354 backup.sql
10/27/2026  10:19 AM            20,377 backup_without_logs.sql
10/27/2026  10:20 AM            26,307 backup_with_ownership.dump
10/27/2026  10:27 AM    <DIR>          base
10/27/2026  10:27 AM    <DIR>          base_wal
10/27/2026  10:12 AM               526 customers.csv
10/27/2026  10:13 AM               526 customers_batch_1.csv
10/27/2026  10:19 AM             8,572 data_only.sql
10/27/2026  10:10 AM               521 employees.csv
10/27/2026  10:11 AM               524 employees.txt
10/27/2026  10:13 AM               285 feedback.csv
10/27/2026  10:19 AM           212,664 full_cluster_backup.sql
10/27/2026  10:19 AM            26,307 full_production_backup.dump
10/27/2026  10:12 AM               237 it_employees.csv
10/27/2026  10:28 AM            26,307 migration.dump
10/27/2026  09:53 AM               165 new_employees.csv
10/27/2026  10:12 AM               413 orders.csv
10/27/2026  10:12 AM               349 products.csv
10/27/2026  09:53 AM               152 products_utf8.csv
10/27/2026  09:53 AM               171 products_with_errors.csv
10/27/2026  09:53 AM               168 sales.csv
10/27/2026  10:19 AM            14,479 schema_only.sql
10/27/2026  10:17 AM             4,690 tables_backup.sql
              22 файлов        392,401 байт
               4 папок  259,979,476,992 байт свободно
```